

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER 2025

Name of the Program: B.Tech

Semester: I

Name of the Course: Fundamental of Computer and Introduction to C Programming

Course Code: **TCS101**

Time: 90 Minutes

Maximum Marks: 50

Note:

- (i) Answer all the questions except Que(2) by choosing **any one of the sub-questions**.
- (ii) All 5 sub-questions in Que(2) are compulsory.
- (iii) Each question carries 10 marks.

Q1	(10 marks)		CO1
(a)	There are different types of memories used in a computer system. Explain each level of the hierarchy (Registers, Cache, Main Memory, Secondary Storage, and Tertiary/Cloud Storage) in terms of speed, cost, and capacity. Why is the hierarchy designed this way instead of having a single large, fast memory?		
OR			
(b)	The execution of a C program involves multiple steps, from writing the code to running the final executable. Draw a neat diagram showing the life cycle of a C program along with all intermediate files (.c, .i, .s, .o, .exe). Explain the role of each stage (preprocessor, compiler, assembler, linker, loader).		
Q2	(10 marks)		CO3
Find the Output with an explanation. Considering int is getting 2 Bytes of space in memory.			
1. #include<stdio.h> void main() { int m = 9, n = 0, p = 0; p += n += m; p += n += m += -p; printf("%d %d %d", m, n, p); }		2. #include<stdio.h> void main() { int a = 15, b = 13, c = 16; int x; x = a - 3 % 2 + c * 2 / '4 % 2 + b / 4; printf("%d", x); }	
3. #include<stdio.h> int main() { int a=0, b=0, x; x=0 && (a b); x = 0 (a == ++b); printf("%d %d %d", a, b, x); return 0; }		4. #include<stdio.h> int main() { int a=0, b=1; int x, c; x = a<b? (a &1): (a 10); c=x ^ b; printf("c= %d x= %d", c, x); return 0; }	
5. #include<stdio.h> int main()			

	<pre>{ int r=69; switch((char) r) { case 'E': case 'I': case 'O': r+=4; } printf("%c", r); return 0; }</pre>		
Q3	(10 marks)		
(a)	What is a flowchart? How flowcharts assist in problem solving phase? Draw a flowchart to accept a 5-digit positive integer number and swap its first and last digit.		
	Sample Input	Sample Output	
	Input a five-digit positive number: 45281	15284	
OR			
(b)	Draw a flowchart which accept a vowel and change it according to following criteria: a will change with e e will change with i i will change with o o will change with u u will change with a		
Q4	(10 marks)		
(a)	Aman trying to check different colors formed with the different values for R, G, B (max=255 min=0). Develop a 'C' code to help him to find the colors formed with different inputted values. • If the value of R, G, B is 255 then display WHITE • If the value of R, G, B is 0 then display BLACK • If the value of R is between 235 to 255 and value of G and B is 0 then display RED • If the value of G is between 255 to 255 and value of R and B is 0 then display GREEN • If the value of B is between 235 and 255 and value of R and G is 0 then display BLUE • If the value of R and G is 255 and value B is 0 then display YELLOW • If the value of R and B is 255 and value G is 0 then display PURPLE • Otherwise display OTHER COLOR		
OR			
(b)	There are some rules to withdraw amount from ATM they are: • Withdrawing amount should be less than balance amount • If your age is less than 18 then you can withdraw maximum 3000 and withdrawing amount should be multiple of 100. • If your age is from 18 to 60 then you can withdraw maximum 50000 and withdrawing amount should be multiple of 500 • If your age is above 60 then you can withdraw maximum 20000 and withdrawing amount should be multiple of 200 For a successful transaction there will be a transaction charge of Rs. 5. Develop a 'C' code input age pf person, balance amount X and withdrawing amount Y. Print the final balance.		
Q5	(10 marks)		
(a)	Explain any two of following with example: 1. Switch-case 2. Qualifiers 3. Unary, binary and ternary operator		
OR			
(b)	Draw a flowchart which accept a positive integer number and print that number that many times.		
	Sample Input	Sample Output	
	Input a positive number: 5	5 5 5 5 5	
	Input a positive number: 3	3 3 3	